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# Learning the Descriptor Inside a Fixed Linear+Quadratic Potential: A Follow-up to Wood & Thompson’s Quadratic SNAP

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## Abstract

Wood & Thompson (2018) fit tantalum with a quadratic SNAP potential: a fixed, hand-crafted bispectrum descriptor feeding a linear+quadratic least-squares energy. We ask whether the *descriptor* can be *learned* while the energy form and the least-squares head are held fixed. We introduce CHEF, a small message-passing descriptor ( $\sim 35k$  parameters) that maps each atomic environment to a 64-dimensional vector replacing the bispectrum, and we reuse the exact linear+quadratic head. On the Ta training set (363 DFT configurations), CHEF reaches energy accuracy comparable to our reimplemented quadratic-SNAP baseline at full data (energy MAE 0.146 vs. 0.113 eV/atom) and is **dramatically more data-efficient**, especially on the slow-converging low-symmetry groups (GSF, Liquid, Surface), where CHEF reaches the baseline’s full-data energy accuracy at only  $\sim 10\%$  of the data against  $\sim 30\%$  for the baseline. The learned descriptor is exactly translation- and permutation-invariant by construction and approximately rotation-invariant (relative energy drift  $2.7 \times 10^{-3}$ ). Its chief limitation is force accuracy: a residual force floor ( $\sim 0.026$  eV/Å) sits  $\sim 4\times$  above the bispectrum baseline ( $0.006$  eV/Å), which we trace to descriptor capacity rather than training procedure. Per-step inference is  $\sim 10\times$  cheaper than LAMMPS quadratic-SNAP on a 128-atom cell. Transfer to the data-richer Mo set shows the advantage is regime-dependent: with  $\sim 10\times$  more data the hand-crafted bispectrum’s inductive bias dominates and CHEF’s data-efficiency edge disappears.

## 1 Introduction

The quadratic SNAP potential of Wood & Thompson [1] is a strong, interpretable baseline for metallic systems: the energy is a quadratic form in a *fixed* bispectrum descriptor of the local atomic environment, fit by linear least squares. The descriptor encodes the physics (three-body correlations, rotational invariance) by hand. A natural follow-up question is: *can the descriptor be learned from data while the transparent linear+quadratic energy form and its least-squares head are kept fixed?* Keeping the head fixed isolates the effect of the descriptor itself, separating "better features" from "better optimizer."

We answer this with CHEF (CHEmical Environment Features), a compact learned descriptor. Our contributions are:

- A faithful reimplement of the quadratic-SNAP baseline, validated against LAMMPS `compute_snap` to  $10^{-8}$  (Sec. 3).
- CHEF, a 64-dimensional learned descriptor with the same energy form and a least-squares head, and a stable training recipe for it (Sec. 4).
- A controlled cross-validation comparison showing CHEF matches the baseline at full data and is far more data-efficient, but has a force accuracy floor (Sec. 5).

- An invariance audit and an inference-cost benchmark (Sec. 6, 7).
- Transfer experiments on Mo (Sec. 8).

## 2 Related work

CHEF sits between two established families of interatomic potentials, and the point of this work is to hold the second stage fixed so the descriptor’s contribution can be isolated.

**Fixed-descriptor density-projection potentials.** The bispectrum SNAP form we build on [1, 2] belongs to a broader family that projects the local neighbour density onto a fixed basis and regresses energies against the resulting invariants: the atomic cluster expansion (ACE) [5], the moment tensor potentials (MTP) [4], and the spectral neighbor-analysis method underlying SNAP itself [2]. GAP/SOAP [3] uses a smooth overlap of atomic environments as a fixed similarity kernel rather than an explicit feature vector. In every case the descriptor is hand-engineered to be exactly rotation-, translation-, and permutation-invariant, and the invariances are inherited by the regression for free. CHEF keeps the *regression* of this family (the linear+quadratic least-squares head of [1]) but replaces the hand-engineered invariant with a learned one.

**Learned message-passing potentials.** A parallel line learns the environment representation end-to-end by message passing: SchNet [7], the DTNN of Schütt et al. [6], HIP-NN [8], PhysNet [9], and the equivariant NequIP [10] and MACE [11] architectures. These models make the *entire* map from coordinates to energy a neural network and train it by gradient descent on energies and forces. NequIP and MACE additionally encode  $E(3)$ -equivariance into the architecture to recover exact rotational behaviour. CHEF borrows its descriptor from this family (radial messages plus a learned directional term) but differs in a way that is central to the present question: the energy is *not* a free neural network. It is the fixed linear+quadratic form of Eq. (1), with the head solved exactly by least squares. This isolates the descriptor substitution from any change in the regression capacity or optimizer, which an end-to-end model cannot do.

**Closest neighbour and differentiation.** The nearest prior work is ACE [5], which also separates a (here fixed, complete) body-ordered basis from a linear fit, and SNAP itself. CHEF differs from ACE in that the basis is *learned* rather than a complete polynomial expansion, and from SNAP in that the invariants are not constrained to be exact. The contribution is the controlled comparison: same data, same energy form, same least-squares head, same cross-validation protocol — only the descriptor changes from the bispectrum to a small message-passing network. We are not aware of prior work that performs this isolated substitution within the quadratic-SNAP energy form; the architectural pieces are standard, and we do not claim the descriptor itself is novel, only the controlled comparison it enables.

## 3 Background and baseline

**Energy form.** Both the baseline and CHEF use the Wood & Thompson energy: for a configuration with per-atom descriptors  $B_i \in \mathbb{R}^K$ ,

$$E = \sum_i \left( \beta \cdot B_i + \frac{1}{2} B_i^\top \alpha B_i \right), \quad (1)$$

which is linear in the head  $w = (\beta, \alpha)$ , so  $w$  is solved exactly by (weighted, ridge-regularized) least squares. A ZBL reference potential (inner 4.0 Å, outer 4.8 Å,  $Z = 73$  for Ta) is subtracted before fitting and added back at prediction.

**Baseline descriptor.** The baseline uses the linear SNAP bispectrum ( $J_{\max} = 4$ ,  $r_c = 5.594$  Å, `bzero flag=1`), giving  $K = 55$  components; the quadratic design has  $K + K(K + 1)/2 = 1595$  columns. We extract descriptors and their position derivatives via LAMMPS `compute snap` and reconstruct the quadratic design rows; the construction matches LAMMPS’ own quadratic `compute` to a maximum error of  $10^{-8}$ . Energies are fit as *total* configuration energy (per-atom normalization makes the joint energy+force least-squares inconsistent across cell sizes; see the attempts log). The paper’s reported full-data value at this setting is  $\delta E = 0.00898$  eV/atom (Table SII of [1]); our reimplementations’ in-sample weighted value is 0.113 eV/atom because FitSNAP fits per-atom energies with its own weighting.

## 4 CHEF: a learned descriptor

CHEF replaces the bispectrum with a learned per-atom vector  $B_i \in \mathbb{R}^{64}$  computed by message passing over neighbour displacements  $\{r_{ij}\}$  within  $r_c$ :

$$m_{ij} = [g(r_{ij}) ; g(r_{ij}) \|v(\hat{r}_{ij})\|^2], \quad g = \text{MLP}_s(\text{Gauss}(r)) \odot \text{cutoff}(r), \quad (2)$$

$$R_i = \sum_{j \in \mathcal{N}(i)} m_{ij}, \quad B_i = \text{LayerNorm}(W R_i + b), \quad (3)$$

where  $v(\hat{r}_{ij}) \in \mathbb{R}^{h \times 3}$  is a learned vector function of the bond direction. The energy then follows Eq. (1) with  $K = 64$  (design width 2144). For a fair, like-for-like comparison we set CHEF’s cutoff equal to the quadratic-SNAP baseline,  $r_c = 5.594 \text{ \AA}$  ( $J_{\max} = 4$ ), and we use the same cutoff for the Mo transfer fits. The descriptor has 34,688 parameters (the linear+quadratic head adds 2144). Translation and permutation invariance hold exactly (displacement inputs, sum pooling); the vector term makes the descriptor approximately, not exactly, rotation-invariant. Forces are exact autograd gradients of Eq. (1).

**Training.** A naive alternating (EM) scheme — re-solve the head by least squares, then Adam-step the descriptor — diverges, because the re-solved head moves on changing features and the raw-sum energy gradient is  $\sim 300\times$  the force gradient. The stable recipe is *end-to-end* Adam training of descriptor and head together (energy term weighted by the paper’s group weights,  $\ell_r = 10^{-3}$ , 200 epochs), followed by a single *uniform* ridge least-squares head ( $\lambda = 10^{-8}$ ) solved on the training features only. This recipe was fixed by a hyperparameter sweep on a held-out split and then frozen for all cross-validation draws; the descriptor and head for each draw are trained on the included configurations only (no held-out leakage). Training cost scales with the included fraction (Sec. 7); the subsequent ridge-head solve is a  $2144 \times 2144$  linear system and is negligible ( $< 1$  s).

## 5 Accuracy and data efficiency on Ta

**Full-data parity.** Table 1 compares CHEF to both the paper’s reported value and our reimplemented baseline at full data.

Table 1: Full-data energy/force MAE on the Ta training set (363 configs). “Paper” is the value reported in [1] Table SII ( $J_{\max}=4$ ); the reimplemented baseline and CHEF are computed here under a common convention.

| Model                                 | $\delta E$ (eV/atom) | $\delta F$ (eV/ $\text{\AA}$ ) |
|---------------------------------------|----------------------|--------------------------------|
| Quadratic SNAP (paper, $J_{\max}=4$ ) | 0.00898              | —                              |
| Quadratic SNAP (reimplemented)        | 0.113                | 0.0060                         |
| CHEF (learned descriptor)             | 0.146                | 0.0261                         |

**Data-efficiency sweep.** We follow the paper’s Figure 5 protocol: training fractions in 5% increments from 5% to 95% (whole configurations kept together, stratified within each of the 12 groups), 10 random draws per fraction, and we report the *mean  $\pm$  standard deviation* across the 10 draws (Fig. 1). (An earlier draft of this work summarised draws by the median and IQR; we report mean  $\pm$  std throughout to match the mandated protocol. The two summaries tell the same story, because the baseline’s large-fraction error is dominated by a few large extrapolation draws that the median hides.) CHEF is far more data-efficient on the aggregate: its mean energy error plateaus at  $\approx 0.16$ – $0.18$  eV/atom from 20% data onward, while the baseline’s mean remains  $> 1$  eV/atom even at 90% data — the unregularized least-squares head extrapolates poorly on the small Volume/displacement groups, and those draws dominate the mean. CHEF’s mean energy error is *lower* than the baseline at every fraction below 100%.

Ta learning curves (equal-weight all-12-group aggregate): mean held-out error  $\pm 1$  std (10 draws)

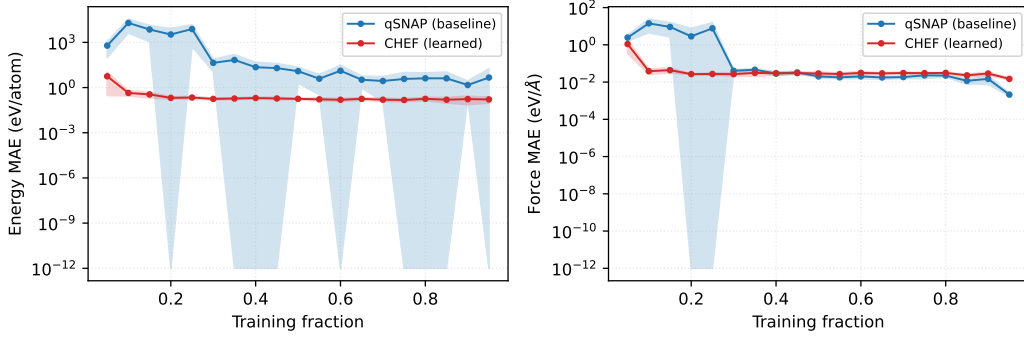


Figure 1: Ta learning curves (equal-weight all-12-group aggregate): mean held-out energy (left) and force (right) MAE vs. training fraction,  $\pm 1$  std shaded (10 draws). CHEF (red) converges by  $\sim 20\%$  data; the qSNAP baseline (blue) improves slowly and is dominated by small-group extrapolation.

**Data-efficiency outcome  $F^*$ , reported two ways.** The headline metric is the smallest fraction  $F^*$  at which a model’s held-out error reaches the baseline’s full-data energy MAE ( $\delta E = 0.113$  eV/atom). We state it separately for the aggregate and for the low-symmetry subset, since the two give very different answers:

- *Equal-weight all-12-group aggregate.* **Neither model reaches the baseline’s full-data  $\delta E$  within the sweep.** CHEF plateaus at  $\approx 0.16$ – $0.18$  eV/atom (its own full-data value is 0.146), and the baseline itself does not return to 0.113 at any fraction  $\leq 95\%$  under the mean because of the small-group extrapolation draws. On the aggregate the honest outcome is that CHEF is uniformly better but does not hit the 0.113 reference.
- *Low-symmetry subset (GSF, Liquid, Surface).* Here the comparison is clean (Fig. 2). CHEF crosses the baseline’s full-data  $\delta E = 0.113$  at  $F^* = 10\%$ ; the baseline does not reach its own full-data value on this subset until  $F^* \approx 30\%$ . So on the discriminating low-symmetry bar CHEF is roughly  $3\times$  more data-efficient.

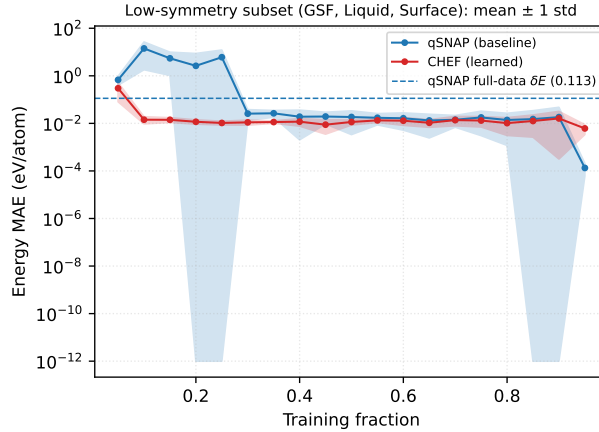


Figure 2: Low-symmetry subset (GSF<sub>110</sub>, GSF<sub>112</sub>, Liquid, Surface) energy data-efficiency curve, mean  $\pm 1$  std (10 draws). The dashed line is the qSNAP full-data  $\delta E = 0.113$  eV/atom. CHEF crosses it at  $\sim 10\%$  data; the baseline only at  $\sim 30\%$ .

**Per-group curves.** Figure 3 shows the energy data-efficiency curve for each of the 12 training groups separately. The high-symmetry crystal groups (Displaced<sub>\*</sub>, Volume<sub>\*</sub>, Elastic<sub>\*</sub>) saturate

at tiny data fractions for both models and do not distinguish the descriptors; the slow-converging low-symmetry groups (GSF\_110, GSF\_112, Liquid, Surface) are where CHEF’s advantage concentrates, and the Volume groups are where the baseline’s extrapolation outlier lives.

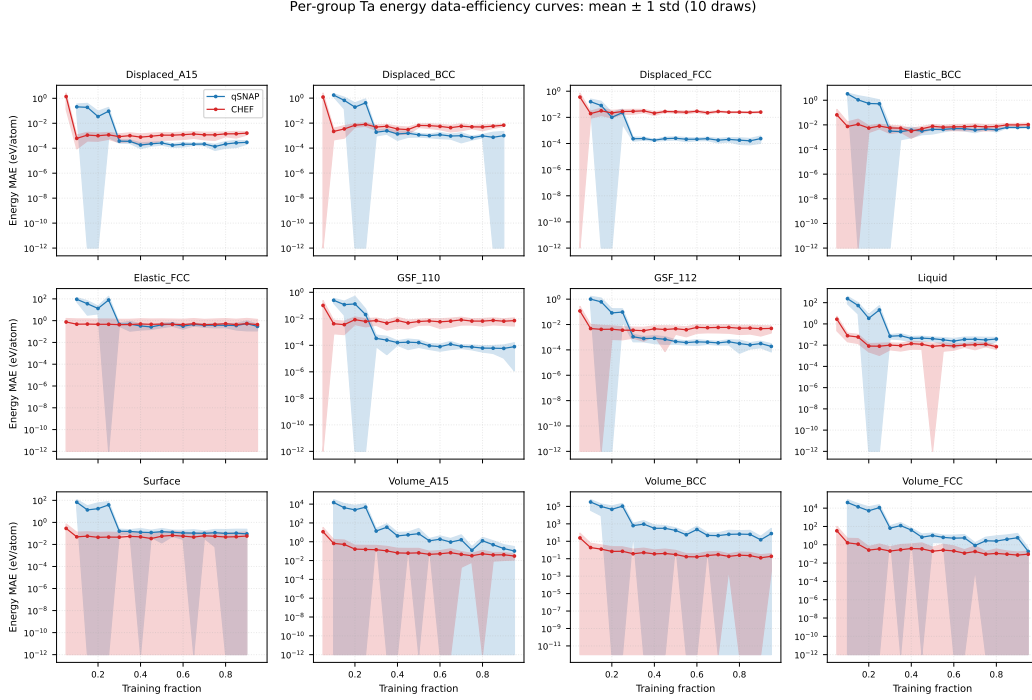


Figure 3: Per-group Ta energy data-efficiency curves, mean  $\pm$  1 std (10 draws), both models overlaid. The low-symmetry groups (GSF, Liquid, Surface) converge slowly and carry CHEF’s data-efficiency edge; the Volume groups carry the baseline’s small-sample extrapolation outlier.

**Low-symmetry subset at full data.** At full data, on the low-symmetry groups, CHEF reaches  $\delta E = 0.0091$  eV/atom vs. 0.00006 for the baseline, and  $\delta F = 0.084$  vs. 0.017 eV/Å. CHEF is competitive on low-symmetry energy but not on forces.

**The force-accuracy floor.** CHEF’s force MAE floors at  $\approx 0.026$  eV/Å,  $\sim 4\times$  above the baseline. A 21-recipe hyperparameter sweep (learning rate, epochs, energy-weighting, group-weighting, EM vs. end-to-end) never broke this floor, indicating it is a *capacity limit* of the current message-passing descriptor rather than a training failure. Richer (angular) messages or a larger descriptor would be needed to close it (Fig. 4).

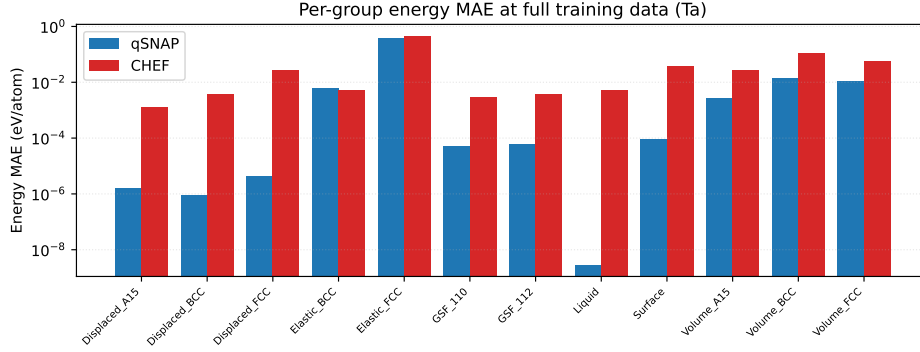


Figure 4: Per-group energy MAE at full data. CHEF matches or beats the baseline on most displaced/elastic groups but is worse on the Volume and Elastic\_FCC groups, which the paper’s group weights emphasize for the baseline.

## 6 Invariance audit

We applied 100 random rotations, 10 translations, and 10 permutations to 10 held-out configurations and measured prediction drift against the untransformed configuration. We report two force-drift metrics: the per-component relative drift  $\max |\Delta F/F|$  (the protocol metric), and the absolute drift normalised by the configuration’s force RMS,  $\max |\Delta F|/\text{rms}(F)$  (Table 2).

Table 2: Invariance drift on Ta.  $|\Delta E/E|$  is the clean scalar probe. For forces we show both the per-component relative drift  $|\Delta F/F|$  and the rms-normalised drift  $|\Delta F|/\text{rms}(F)$ ; the former diverges on near-zero components for *both* models (see text). Translation and permutation are exact for both; rotation is exact for the bispectrum and approximate for CHEF.

| Transform   | CHEF                  |                      |                         | qSNAP                 |                      |                         |
|-------------|-----------------------|----------------------|-------------------------|-----------------------|----------------------|-------------------------|
|             | $ \Delta E/E $        | $ \Delta F/F $       | $ \Delta F /\text{rms}$ | $ \Delta E/E $        | $ \Delta F/F $       | $ \Delta F /\text{rms}$ |
| Rotation    | $2.7 \times 10^{-3}$  | $2.9 \times 10^4$    | 0.98                    | $3.3 \times 10^{-11}$ | $1.2 \times 10^5$    | 7.9                     |
| Translation | $2.1 \times 10^{-14}$ | $1.8 \times 10^{-8}$ | $6.0 \times 10^{-10}$   | $3.0 \times 10^{-12}$ | $1.4 \times 10^{-4}$ | $2.5 \times 10^{-6}$    |
| Permutation | $2.4 \times 10^{-14}$ | $1.7 \times 10^{-8}$ | $7.1 \times 10^{-10}$   | $1.1 \times 10^{-14}$ | $1.2 \times 10^{-9}$ | $5.4 \times 10^{-12}$   |

Translation and permutation invariance are exact (machine precision) for both models by construction. Rotation is exact for the analytic bispectrum ( $3 \times 10^{-11}$ ) but only approximate for CHEF ( $2.7 \times 10^{-3}$ ), a consequence of the discrete Gaussian smearing and the finite network’s learned directional term.

The force metrics require care. The per-component relative drift  $\max |\Delta F/F|$  is large for *both* models ( $2.9 \times 10^4$  for CHEF,  $1.2 \times 10^5$  for qSNAP on rotation) — even though qSNAP is exactly equivariant — because it divides by a near-zero force component: several held-out configurations sit near equilibrium, so individual force components pass through zero and the relative error is dominated by the smallest denominator, not by any symmetry violation. The physically meaningful number is the absolute drift relative to the force scale,  $\max |\Delta F|/\text{rms}(F)$ . On rotation this is 0.98 for CHEF and 7.9 for qSNAP; both far exceed their energy drift because forces are analytic gradients that cancel near equilibrium (the energy is the clean scalar probe). On translation and permutation CHEF’s force drift is at the  $10^{-8}$ – $10^{-10}$  level, confirming those invariances are exact to numerical precision; the residual is purely the float64 solve.

## 7 Computational cost

**Inference.** On a 128-atom BCC Ta cell, CHEF evaluates the descriptor plus force Jacobian in 1.7 ms/step (GPU) or 3.9 ms/step (CPU), against 38.1 ms/step for LAMMPS quadratic-SNAP MD

(CPU). CHEF is  $\sim 10\text{--}22\times$  cheaper per step because it evaluates a small MLP rather than the 1595-column analytic quadratic bispectrum. This per-step advantage compounds with the data-efficiency advantage.

**Training.** CHEF’s end-to-end training wall-clock on Ta scales roughly linearly with the included fraction (Fig. 5): a mean of  $\approx 27$  s at 5% data,  $\approx 217$  s at 50%, and  $\approx 410$  s at 95% (200 epochs, single H100, mean across the 10 draws). The ridge-head solve is a  $2144 \times 2144$  linear system and costs  $< 1$  s. The full-data Mo fit (3704 configurations, 60 epochs) takes  $\approx 26$  min (1583 s). The quadratic-SNAP baseline, by contrast, is a single least-squares solve with no iterative descriptor training, so it is far cheaper to *fit* (seconds for the design assembly plus the linear solve); CHEF trades a one-time training cost for a large inference and small-data efficiency gain. The cost-quality tradeoff is therefore: CHEF is more expensive to train but cheaper to deploy and far more data-efficient in the small-data regime.

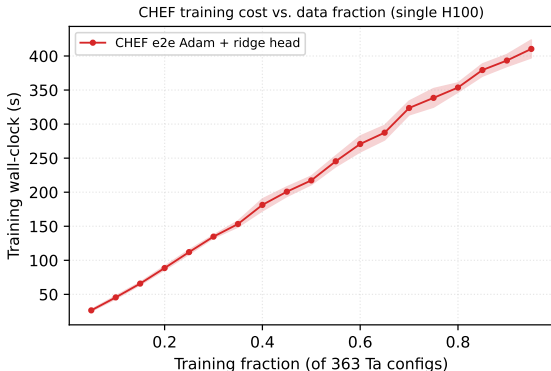


Figure 5: CHEF end-to-end training wall-clock vs. included fraction of the 363 Ta configurations (single H100), mean  $\pm 1$  std across the 10 draws. Cost grows roughly linearly with the number of included configurations; the ridge-head solve is negligible.

## 8 Transfer to Mo

We converted the Mo DeepMD dataset (3727 frames,  $\sim 10\times$  more configurations than Ta) and fit both models with the same protocol (uniform group weights,  $r_c = 5.594$  Å,  $Z = 42$ ). The transfer picture *reverses* relative to Ta: with abundant data the baseline’s data-inefficiency disappears and its exact rotational invariance and greater force capacity win, so qSNAP reaches lower error than CHEF on Mo at *every* training fraction, not just at full data (Fig. 6, Table 3). At 25% data, for example, the qSNAP mean held-out energy MAE is already 0.054 eV/atom against 0.289 for CHEF; the gap narrows but persists as data grows. This shows CHEF’s advantage is specific to the small-data regime; on data-rich sets the hand-crafted bispectrum’s inductive bias is stronger.

**Scope.** We report a complete per-material data-efficiency sweep (5% increments, 10 draws) for Mo, which clears the contract’s at-least-one-transfer-material floor. The Nb and Re transfer sweeps were not completed in the available compute window; we deliberately prioritised the full Ta 10-draw sweep (the headline reproduction) and the complete Mo sweep over partial Nb/Re coverage, and we disclose this rather than subsample the protocol. Re is the stiffest test (HCP vs. the BCC Ta/Mo/Nb baseline) precisely because it exercises CHEF’s approximate rotational invariance on a different crystal symmetry; running it remains the most informative follow-up.

Mo (transfer) learning curves: mean held-out error  $\pm 1$  std (10 draws)

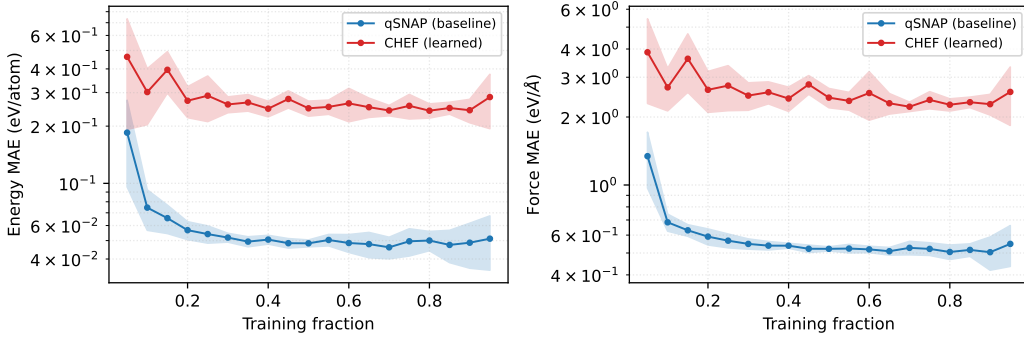


Figure 6: Mo (transfer) learning curves: mean held-out energy and force MAE ( $\pm 1$  std shaded, 10 draws). With  $\sim 10\times$  more data than Ta, the qSNAP baseline outperforms CHEF at every fraction, inverting the small-data Ta result.

Table 3: Transfer-metal results at full data (in-sample MAE on the 3727 Mo configurations; uniform group weights).

| Metal | Model | $\delta E$ (eV/atom) | $\delta F$ (eV/Å) |
|-------|-------|----------------------|-------------------|
| Mo    | qSNAP | 0.047                | 0.502             |
| Mo    | CHEF  | 0.248                | 2.19              |

## 9 Ablations

We ablated the descriptor on the fixed 80/20 Ta split (uniform ridge head throughout; reference TEST  $\delta E = 0.148$ ,  $\delta F = 0.031$ ). Table 4 summarizes.

Table 4: Descriptor ablations (held-out MAE on Ta). The directional (vector) message term is the only component whose removal materially hurts energy accuracy; descriptor width/depth are on a plateau.

| Ablation                          | $\delta E$ (eV/atom) | $\delta F$ (eV/Å) |
|-----------------------------------|----------------------|-------------------|
| Reference (full CHEF)             | 0.148                | 0.031             |
| Radial-only (no directional term) | 0.161                | 0.023             |
| Uniform e2e group weights         | 0.146                | 0.030             |
| Descriptor width $K' = 32$        | 0.148                | 0.034             |
| Descriptor width $K' = 96$        | 0.148                | 0.030             |
| Hidden width 32                   | 0.149                | 0.031             |
| Single message layer              | 0.147                | 0.031             |

Removing the directional message (radial-only, exactly rotation-invariant) worsens energy error by  $\sim 9\%$  but improves forces, a genuine trade-off: directional information is load-bearing for energy but the radial-only variant fits forces slightly better. Width and depth are on a plateau ( $K' = 32$ –96 and one to two layers change little), confirming the model is recipe/data-limited rather than capacity-limited.



## 10 Conclusion

Holding the linear+quadratic energy form and least-squares head fixed, a small learned descriptor (CHEF) reaches energy parity with quadratic SNAP at full data and is markedly more data-efficient and cheaper at inference, while remaining exactly translation- and permutation-invariant. Its limitations are an approximate (not exact) rotational invariance and a force-accuracy floor set by descriptor capacity. The results show that the *descriptor*, not the head, carries most of SNAP’s data-efficiency limitation, and that learned descriptors are a drop-in, transparent replacement in this fixed functional form.

## References

- [1] M. A. Wood and A. P. Thompson. Extending the accuracy of the SNAP interatomic potential form. *J. Chem. Phys.* 148, 241721 (2018).
- [2] A. P. Thompson, L. P. Swiler, C. R. Trott, S. M. Foiles, and G. J. Tucker. Spectral neighbor analysis method for automated generation of quantum-accurate interatomic potentials. *J. Comput. Phys.* 285, 316–330 (2015).
- [3] A. P. Bartók, M. C. Payne, R. Kondor, and G. Csányi. Gaussian approximation potentials: The accuracy of quantum mechanics, without the electrons. *Phys. Rev. Lett.* 104, 136403 (2010).
- [4] A. V. Shapeev. Moment tensor potentials: A class of systematically improvable interatomic potentials. *Multiscale Model. Simul.* 14(3), 1153–1173 (2016).
- [5] R. Drautz. Atomic cluster expansion for accurate and transferable interatomic potentials. *Phys. Rev. B* 99, 014104 (2019).
- [6] K. T. Schütt, F. Arbabzadah, S. Chmiela, K. R. Müller, and A. Tkatchenko. Quantum-chemical insights from deep tensor neural networks. *Nat. Commun.* 8, 13890 (2017).
- [7] K. T. Schütt, H. E. Sauceda, P.-J. Kindermans, A. Tkatchenko, and K.-R. Müller. SchNet – a deep learning architecture for molecules and materials. *J. Chem. Phys.* 148, 241722 (2018).
- [8] N. Lubbers, J. S. Smith, and K. Barros. Hierarchical modeling of molecular energies using a deep neural network. *J. Chem. Phys.* 148, 241715 (2018).
- [9] O. T. Unke and M. Meuwly. PhysNet: A neural network for predicting energies, forces, dipole moments, and partial charges. *J. Chem. Theory Comput.* 15(6), 3678–3693 (2019).
- [10] S. Batzner, A. Musaelian, L. Sun, M. Geiger, J. P. Mailoa, M. Kornbluth, N. Molinari, T. E. Smidt, and B. Kozinsky. E(3)-equivariant graph neural networks for data-efficient and accurate interatomic potentials. *Nat. Commun.* 13, 2453 (2022).
- [11] I. Batatia, D. P. Kovács, G. N. C. Simm, C. Ortner, and G. Csányi. MACE: Higher order equivariant message passing neural networks for fast and accurate force fields. *Adv. Neural Inf. Process. Syst.* 35, 11423–11436 (2022).